MIT WORLD PEACE UNIVERSITY

SCHOOL OF SCIENCE AND ENVIRONMENTAL STUDIES

DEPARTMENT OF MATHEMATICS AND STATISTICS

A PROJECT REPORT ON

IMPACT OF AI AWARENESS ON CAREER ANXIETY AMONG STUDENTS

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ABSTRACT

This study examined the relationship between students' awareness of Artificial Intelligence (AI) and their career-related anxiety. A cross-sectional survey of [N = 102] students was conducted using a structured questionnaire measuring AI awareness (3 Questions) and career anxiety (4 Questions), along with demographic items (age, stream, education).

Data were analysed using SPSS and EXCEL to compute descriptive statistics, reliability (Cronbach's alpha), correlation, and group comparisons (ANOVA / chi-square as applicable).

Results showed that mean AI awareness was M = [3.7941], SD = [.83783], and mean career anxiety was M = [2.9118], SD = [.88974].

A Pearson correlation indicated a no relationship between awareness and anxiety ($r = .072$, $p = 0.474$)

ANOVA revealed a significant difference in awareness across education levels ($F = 3.008$, $p = .022$).

These findings suggest Institutions should focus on AI education, skill-building, and career guidance to help students manage anxiety and adapt to AI-related changes effectively.

INTRODUCTION

Artificial Intelligence (AI) is rapidly transforming workplaces and professional roles across sectors. As AI tools become more accessible and powerful, students face both opportunities such as new jobs and efficiency gains and uncertainties about future career prospects. Understanding how students' awareness of AI relates to their career anxiety is important for designing curriculum interventions and career guidance that reduce unnecessary fear while promoting adaptive skill-building.

This study investigates the relationship between AI awareness and career anxiety among students.

Specifically, it asks: does greater awareness of AI associate with higher or lower career anxiety?

The study also explores whether awareness differs by demographic factors such as education level, age group, and academic stream.

OBJECTIVES

1. To measure the level of AI awareness among students.
2. To measure the level of career-related anxiety associated with AI among students.
3. To examine the relationship between AI awareness and career anxiety.
4. To investigate differences in AI awareness and career anxiety across education levels, age groups, and academic streams.

METHODOLOGY

Research design

A cross-sectional survey design was used to collect data from students.

Participants

The study sample comprised [N = 102] students who completed the questionnaire. Participants came from various age groups and academic streams.

Survey

Data were collected using a structured questionnaire with three sections:

- **Section A: Demographics -**

Age -

[16-20, 21 – 24, 25 – 30, 30 above]

What is your academic stream?

[Science, Commerce, Arts/Humanities, Management, Engineering, Other]

Education –

[High School, UG, PG, Diploma, Other]

- **Section B: AI Awareness –**

1) How familiar are you with Artificial Intelligence (AI)?

2) How often do you use AI tools (e.g. ChatGPT, Google Gemini, AI apps)?

3) How confident are you in your ability to adapt to AI-related changes in your career?

- **Section C: Career Anxiety –**

4) Does the rapid growth of AI make you anxious about your future career?

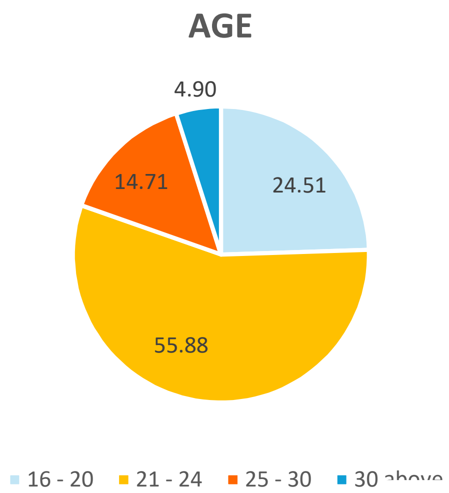
5) Do you feel that AI may reduce job opportunities in your chosen stream?

6) Do you feel pressure to learn AI-related skills to remain employable?

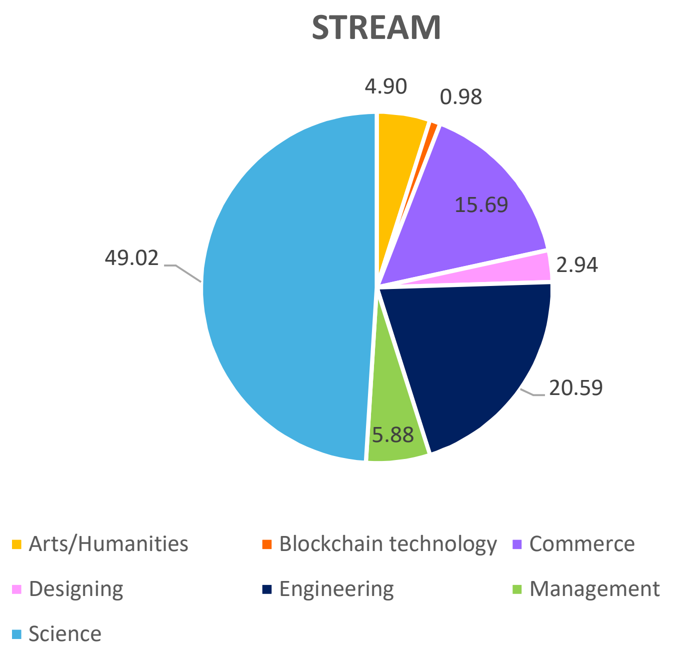
7) Overall, how high is your career anxiety due to AI awareness?

EXPLORATORY DATA ANALYSIS

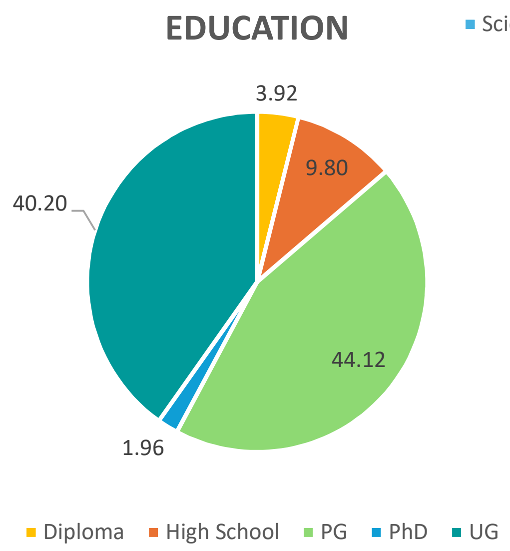
DEMOGRAPHICS



In our sample, 55.88% of students were from 21 – 24 age group and 24.51% of students from 16 – 20 age group.

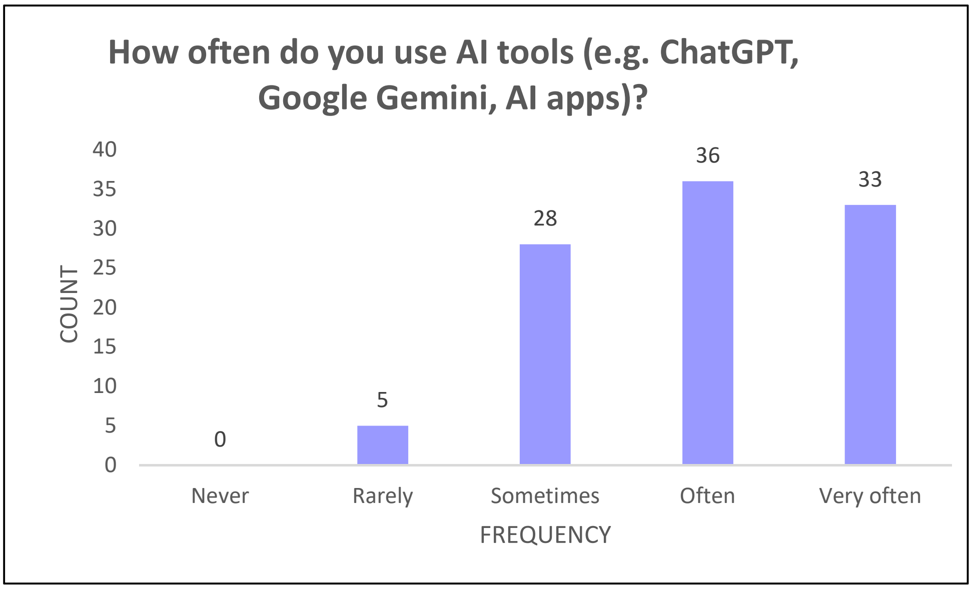


In our sample, 49.02% of students were from "SCIENCE" stream and 20.59% of students from "ENGINEERING" stream.

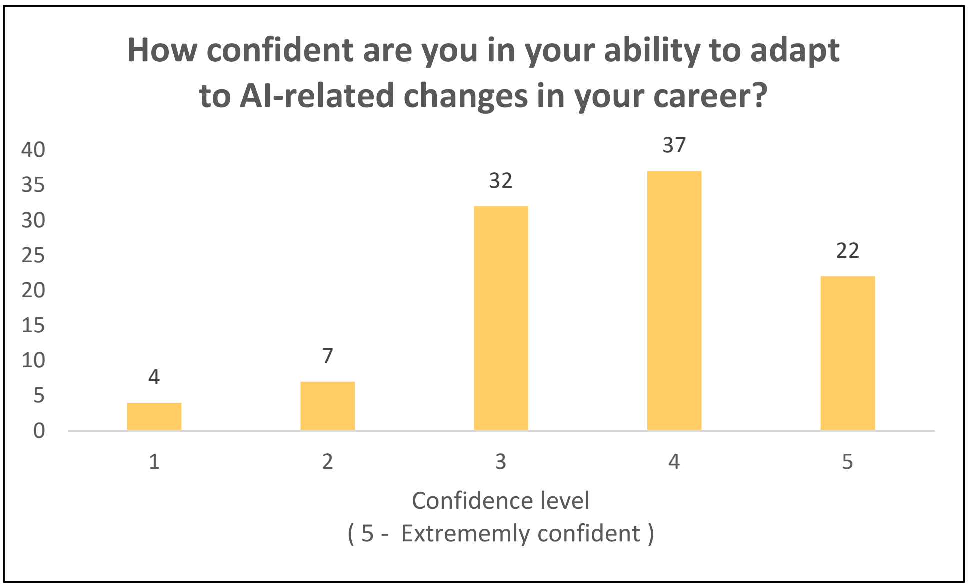


In our sample, 44.20% of students were from PG and 40.20% of students from UG.

AI AWARENESS

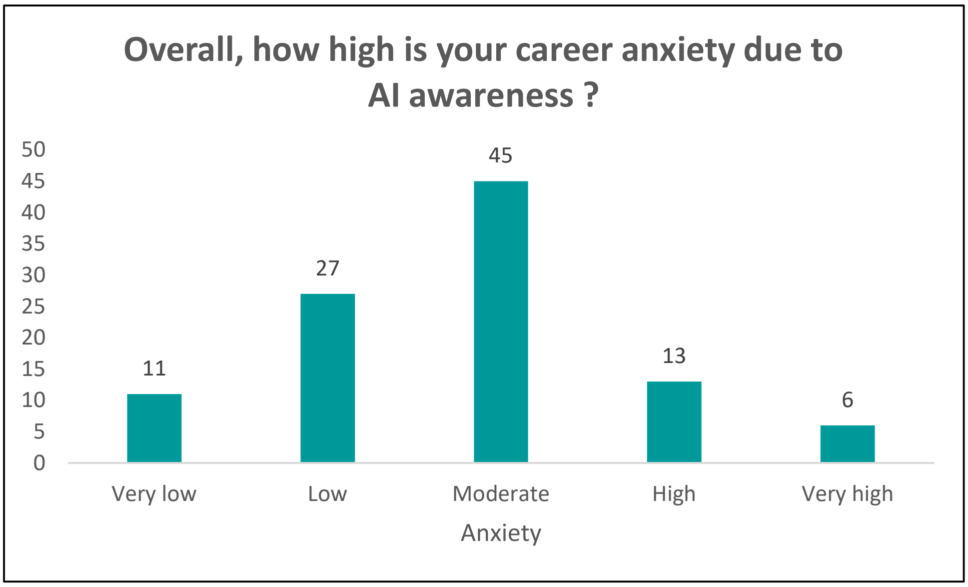


Most of the students use AI tools more often.

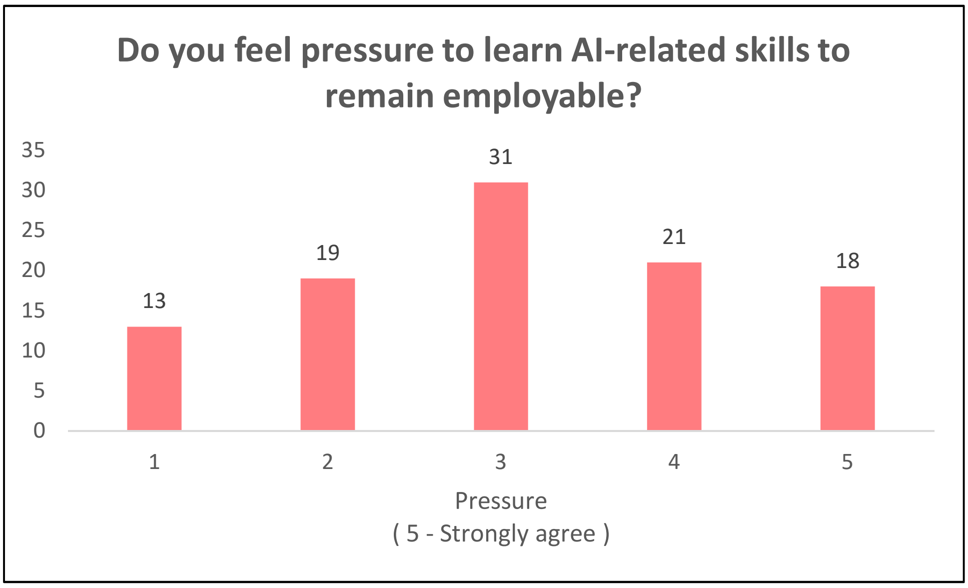


Students are confident that they can adapt easily to AI-related changes in their career.

CAREER ANXIETY



AI awareness generates the moderate career anxiety among the surveyed population/students.



The majority of students feel a moderate to high level of pressure to learn AI-related skills to remain employable.

ANALYSIS

A. RELIABILITY:

The reliability of the scale was assessed using Cronbach's Alpha and the values obtained are:

SECTION A: 0.735

SECTION B: 0.762

Hence, we can say that our data is reliable.

B. DESCRIPTIVE:

Overall anxiety and awareness scores

		Awareness	Anxiety
N	Valid	102	102
	Missing	0	0
Mean		3.7941	2.9118
Median		4.0000	3.0000
Std. Deviation		.83783	.88974
Variance		.702	.792

AI AWARENESS

1) Age

Age	Average of How familiar are you with Artificial Intelligence (AI)?
16 - 20	3.56
21 - 24	4.01754386
25 - 30	3.6
30 above	2.8

Higher familiarity among 21–24-year-olds may contribute to their slightly higher career anxiety.

2) Education

Education	Average of How familiar are you with Artificial Intelligence (AI)?
Diploma	2.25
High School	3.1
PG	3.777777778
PhD	4.5
UG	4.073170732

AI familiarity tends to increase with higher education levels, with PhD and UG students being the most familiar.

3)Stream

What is your academic stream?	Average of How familiar are you with Artificial Intelligence (AI)?
Arts/Humanities	4
Commerce	3.125
Designing	3.333333333
Engineering	4.428571429
Management	3.833333333
Science	3.7

Engineering students being the most familiar with AI.

CAREER ANXIETY

1) Age

Age	Average of Overall, how high is your career anxiety due to AI awareness?
16 - 20	2.56
21 - 24	2.859649
25 - 30	2.8
30 above	2.6
Grand Total	2.764706

Career anxiety related to AI is roughly the same across all age groups.

2) Education

Education	Average of Overall, how high is your career anxiety due to AI awareness?
Diploma	2.25
High School	2.3
PG	3.155556
PhD	2.5
UG	2.512195
Grand Total	2.764706

The highest average career anxiety is observed among Postgraduate (PG) students 3.15.

3) Stream

Stream	Average of Overall, how high is your career anxiety due to AI awareness?
Arts/Humanities	2.4
Commerce	2.875
Designing	2.666667
Engineering	2.714286
Management	2.833333
Science	2.78
Grand Total	2.764706

The average career anxiety is moderate across all stream.

C. ANOVA:

- $p < 0.05$:
  The result is considered statistically significant.
  This suggests the relationship or difference observed in the data is unlikely to be due to chance.
- $p \geq 0.05$:
  The result is not statistically significant, meaning the data does not provide strong evidence against the null hypothesis.

1) Ho: There is no significant difference in AI Awareness across different education levels.

H1: There is a significant difference in AI Awareness across different education levels.

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	7.824	4	1.956	3.008	.022
Within Groups	63.075	97	.650		
Total	70.899	101			

Interpretation: Here $p = 0.022 < 0.05$

Hence, we reject Ho.

2) Ho: There is no significant difference in mean anxiety level across different education levels.

H1: There is significant difference in mean anxiety level across different education levels.

Interpretation: Here $p = 0.056 >= 0.05$

Hence, we accept Ho.

3) Ho: There is no significant difference in awareness across the age groups.

H1: There is significant difference in awareness across the age groups.

Interpretation: Here $p = 0.015 < 0.05$

Hence, we reject Ho.

4) Ho: There is no significant difference in anxiety levels among the age groups.

H1: There is significant difference in anxiety levels among the age groups.

Interpretation: Here $p = 0.912 > 0.05$

Hence, we accept Ho.

D. CHI SQUARE TEST:

1) Ho: There is no significant association between age group and confidence in adapting to AI-related career changes.

H1: There is a significant association between age group and confidence in adapting to AI-related career changes.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	25.199[a]	12	.014
Likelihood Ratio	23.443	12	.024
Linear-by-Linear Association	.144	1	.704
N of Valid Cases	102		

a. 13 cells (65.0%) have expected count less than 5. The minimum expected count is .20.

Interpretation:

The p-value = 0.014 < 0.05, then Ho is rejected.

2) Ho: There is no significant relationship between confidence in adapting to AI-related changes and the pressure to learn AI-related skills.

H1: There is significant relationship between confidence in adapting to AI-related changes and the pressure to learn AI-related skills.

Interpretation:

The p-value =0.012< 0.05, then Ho is rejected.

3)Ho: There is no significant association between how often individuals use AI tools and the level of anxiety they feel about their future career due to AI growth.

H1: There is significant association between how often individuals use AI tools and the level of anxiety they feel about their future career due to AI growth.

Interpretation:

The p-value = 0.579 > 0.05, then Ho is accepted.

4) Ho: There is no significant association between familiarity with AI and career anxiety due to AI awareness.

H1: There is significant association between familiarity with AI and career anxiety due to AI awareness.

Interpretation:

The p-value = 0.423 > 0.05, then Ho is accepted.

E. CORRELATION:

Ho: There is no significant correlation between AI Awareness and Career Anxiety among students.

H1: There is a significant correlation between AI Awareness and Career Anxiety among students.

		Awareness	Anxiety
Awareness	Pearson Correlation	1	.072
	Sig. (2-tailed)		.474
	N	102	102
Anxiety	Pearson Correlation	.072	1
	Sig. (2-tailed)	.474	
	N	102	102

Interpretation:

The Pearson correlation between AI Awareness and Career Anxiety was found to be weak and statistically insignificant ($r = 0.072$, $p > 0.05$). This indicates that AI awareness does not meaningfully predict or influence career anxiety among the students in the sample. In other words, students with higher awareness of AI do not necessarily experience higher or lower career anxiety.

RESULT

1. AI Awareness

- Overall mean awareness: 3.79 (moderately high).
- Significant differences in AI awareness were observed across education levels ($p = 0.022$) and age groups ($p = 0.015$).
- Higher awareness observed among 21–24-year-olds, UG and PhD students, and Engineering stream students.

2. Career Anxiety

- Overall mean anxiety: 2.91 (moderate).
- ANOVA results: No significant difference in anxiety across age groups ($p = 0.912$) or education levels ($p = 0.056$).
- Slightly higher anxiety in Postgraduate students (3.16) and 21–24-year-olds (2.86).
- Across streams, moderate anxiety was observed, highest in Commerce (2.88) and Management (2.83).

3. Chi-Square Findings

- Significant associations:
    - Age and confidence to adapt to AI changes ($p = 0.014$)

    - Confidence in adapting and pressure to learn AI skills ($p = 0.012$)
- No significant associations:
    - AI tool usage and career anxiety ($p = 0.579$)
    - AI familiarity and career anxiety ($p = 0.423$)

4. Correlation Analysis

- AI awareness and career anxiety showed a weak, non-significant correlation ($r = 0.072$, $p = 0.474$).

IMPLICATIONS

1. **Educational Programs:** AI awareness varies by age and education, so targeted AI education and early exposure are needed, especially for younger and less-educated students.
2. **Career Guidance:** Since confidence to adapt to AI impacts perceived pressure to learn skills, career counselling should focus on building adaptability and coping strategies.
3. **Skill Development:** Workshops and training should enhance both AI-related skills and confidence in using technology, reducing stress associated with skill acquisition.
4. **Policy and Curriculum:** Institutions should integrate structured AI learning and adaptability programs into curricula to prepare students for AI-driven careers.

CONCLUSION

The present study examined the impact of AI awareness on career anxiety among students and found no statistically significant correlation between the two variables. This indicates that students' levels of awareness or understanding of artificial intelligence do not directly influence how anxious they feel about their future careers. Career anxiety appears to be shaped more by broader personal, academic, and social factors rather than by AI-related knowledge alone. These results challenge the common assumption that greater exposure to AI automatically reduces or increases anxiety and instead highlight the complexity of students' psychological responses to technological change.

Despite the absence of a direct relationship between AI awareness and career anxiety, the study identified significant associations involving key demographic variables. Age and education level were found to have a significant relationship with AI awareness, with older students and those pursuing higher educational qualifications demonstrating greater familiarity with AI concepts and technologies. This suggests that exposure to advanced coursework, increased academic engagement, and greater maturity contribute to a deeper understanding of AI and its implications.The analysis further revealed that age is significantly associated with confidence in adapting to AI-related career changes. Older students tend to report higher adaptability confidence, possibly due to more experience, better self-efficacy, or greater technological exposure during their academic journey.

Additionally, the study found a significant relationship between students' confidence in adapting to AI-driven changes and the perceived pressure to learn AI-related skills**.** Students with lower adaptability confidence tend to experience greater pressure to upskill, whereas those who feel more capable of adjusting to technological advancements report lower levels of such pressure. This relationship highlights the psychological interplay between self-efficacy and skill-related stress in the context of emerging AI demands.

Overall, the findings suggest that while AI awareness does not directly influence career anxiety, related factors—particularly age, education level, adaptability confidence, and skill-pressure—play an important role in shaping students' perceptions of AI within their academic and career development pathways. These insights point to the importance of targeted AI education, early technological exposure, confidence-building interventions, and comprehensive career counselling. By addressing these areas, educational institutions can better equip students to navigate an AI-driven labor market with resilience, clarity, and informed perspectives.

-: THANK YOU :-